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# EVORA LABS CUSTOMER SPENDING PERFORMANCE ANALYSIS

Understanding Revenue Drivers  
Through Customer Behavior

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**EVA ARLIANA**

DEEPP Final Project

[LinkedIn](#) | [GitHub](#) | [Dashboard Link](#)



# Executive Summary

## Business Problem

Evora Labs achieved strong revenue performance, generating **\$33.97M** with **8.9%** growth. However, only **26.3%** of customers made repeat purchases, while 38.3% have never made a purchase, indicating opportunities to improve customer activation and retention.

## Key Findings

💰 Revenue reached \$33.97M.

📈 Revenue Growth reached 8.9%.

🔄 Only 26.3% of customers made repeat purchases.

👥 Senior customers contribute highest revenue.

🎁 Jacket is top-performing product.

📍 Queensland leads revenue contribution.

## Recommendations

✓ Improve customer activation & retention.

✓ Target high-value segments.

✓ Prioritize top products

✓ Expand regional campaigns.

# Project Background



## Company

Evora Labs is an online retail company that relies on customer purchasing activity to drive revenue growth.



## Dataset

- 1,000 Customers
- 1,000 Orders
- 5,000 Sales Records
- 1,260 Products
- Time Period: 2021



## Context

Analyze customer purchasing behavior to identify key drivers of revenue performance and uncover opportunities for sustainable growth.

Region: Australia

Industry: E-Commerce

Business Model: B2C

# Business Questions & Objectives



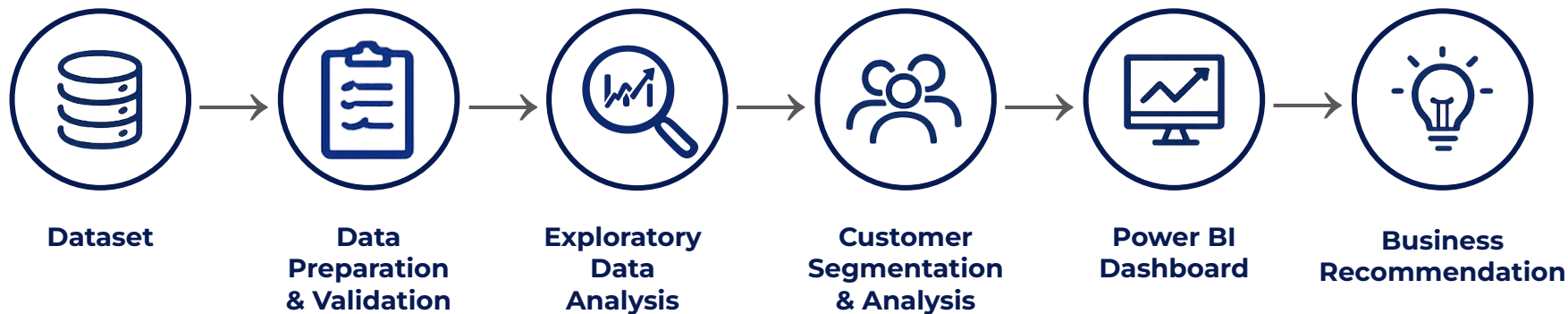
## Business Question

- ? What factors drive customer spending performance?
- ? Which customer segments contribute the highest revenue?
- ? How do purchase frequency and average order value impact revenue?
- ? Which products and regions generate the strongest performance?

## Objectives

- ✓ Analyze customer spending behavior
- ✓ Measure purchase frequency and repeat rate
- ✓ Identify high-value customer segments
- ✓ Evaluate product and regional performance
- ✓ Generate actionable business recommendations

# Methodology



## Tools Used



BigQuery SQL



Power BI

# Data Analysis (Key Findings)



## Customer Retention

Customer activation and retention remain key growth opportunities, with only **26.3%** repeat customers.



## Customer Concentration

Revenue is concentrated among a relatively small group of customers, creating dependency on high-value buyers.



## Customer Demographics

**Senior** customers contribute the highest revenue and represent the most valuable customer segment.



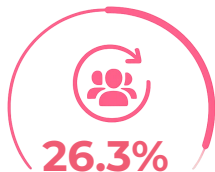
## Product & Regional Performance

**Jackets** and **Queensland** are the strongest revenue contributors.

# Insight 1: Customer Retention Gap

*A large share of customers remain inactive or purchase only once.*

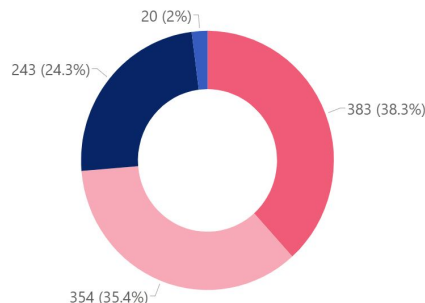
## Repeat Rate



Of customers make repeat purchases

**Below 30%**  
Improvement Opportunity

## Purchase Frequency Distribution



### Purchase Segment

Customers % of Total

|              |     |       |
|--------------|-----|-------|
| No Purchase  | 383 | 38.3% |
| 1 Purchase   | 354 | 35.4% |
| 2-3 Purchase | 243 | 24.3% |
| 4+ Purchase  | 20  | 2%    |



## Key Finding

Only **26.3%** of customers made **repeat purchases**, while **38.3%** have **never made a purchase**. This highlights opportunities to improve customer activation and retention.



## Business Impact

Revenue growth depends heavily on activating inactive customers and increasing repeat purchases among existing customers.



## Recommendation

Implement customer activation campaigns, loyalty programs, personalized promotions, and post-purchase engagement strategies.



**Key Takeaway :** Improving customer activation and retention represents the largest opportunity to drive sustainable revenue growth.

# Insight 2: Senior Customer Drive Revenue

Senior customers contribute the highest revenue and customer base among all age groups.



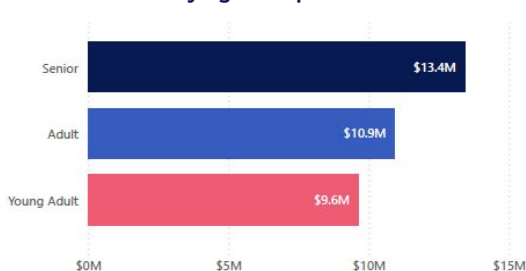
Most Valuable Segment  
**Senior Customers**  
(>55 years old)

**\$13.4M**  
Revenue

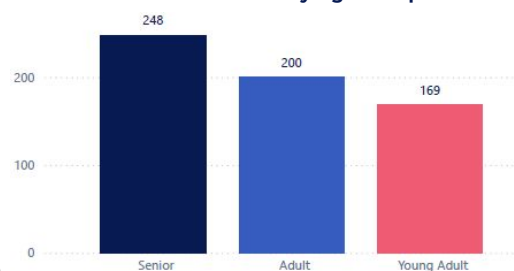


**248**  
Customer

Revenue by Age Group



Customer Distribution by Age Group



## Key Finding

Senior customers generate the highest revenue (**\$13.4M**) and represent the largest customer segment (**248 customers**), making them the most valuable customer group.



## Business Impact

Business performance is highly influenced by senior customers, indicating significant revenue opportunities through targeted engagement and retention efforts.



## Recommendation

Develop personalized promotions, loyalty programs, and retention campaigns tailored to senior customers to maximize customer lifetime value.



**Key Takeaway:** Senior customers are the most valuable customer segment and should be prioritized in retention and marketing strategies.





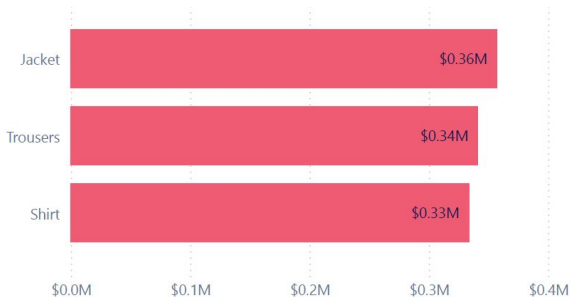
# Insight 3: Top Products & Regions

*Jackets and Queensland lead revenue performance.*



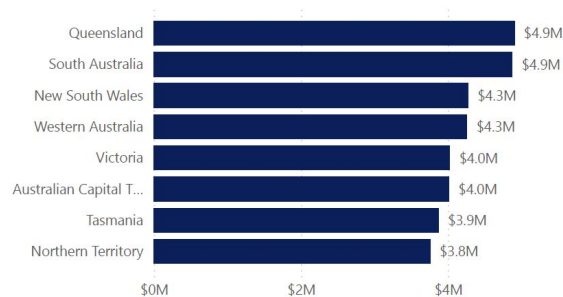
**Revenue by Product Type**  
(in Million USD)

★ Top Product  
**Jacket**  
**\$0.36M**



**Revenue by State**  
(in Million USD)

🏆 Top Region  
**Queensland**  
**\$4.9M**



## Business Impact

Revenue performance is concentrated in a limited number of products and regions, creating opportunities to maximize growth through focused commercial strategies.



## Recommendation

Prioritize inventory planning, promotional campaigns, and regional marketing investments in top-performing products and regions.



**Key Takeaway :** High-performing products and regions should be prioritized to maximize revenue growth and improve resource allocation efficiency.



# Insight 4: Revenue Growth Remains Volatile

Monthly revenue fluctuates despite achieving positive overall growth.



Revenue Growth  
**8.9%**  
Overall Growth



Peak Revenue  
**\$4.1M**  
March

## Revenue Trend by Month



## Key Finding

Revenue reached a peak of **\$4.1M in March**, but monthly performance fluctuated throughout the year despite achieving an overall growth rate of **8.9%**.



## Business Impact

Revenue volatility may indicate seasonal purchasing behavior and inconsistent customer activity across different periods.



## Recommendation

Identify seasonal trends and strengthen retention or promotional initiatives during lower-performing months to stabilize revenue performance.



Key Takeaway: Sustainable growth requires reducing revenue volatility and maintaining consistent customer purchasing activity throughout the year.



# Dashboard Overview

Power BI Recording

## 1 KPI Summary

- Total Revenue
- Total Customers
- Average Order Value
- Repeat Rate
- Revenue Growth %

## 2 Revenue Trend

- Monthly Revenue
- Growth Trend

## 3 Customer Insights

- Purchase Frequency Distribution
- Customer Distribution by Age Group
- Revenue by Age Group

## 4 Product Performance

- Revenue by Product Type

## 5 Geographic Performance

- Revenue by State

## EVORA LABS CUSTOMER SPENDING PERFORMANCE DASHBOARD

Understanding Revenue Drivers Through Customer Behavior

State: All | Month: All | Gender: All | Age: All | Product: All

Total Revenue

\$33.97M

Revenue Growth %

8.9%

Total Customers

1K

Average Order Value

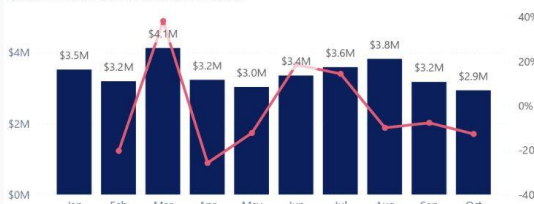
\$33.97K

Repeat Rate

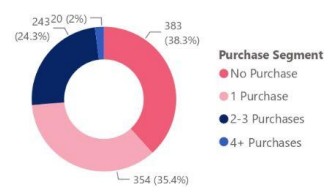
26.3%

Revenue Trend by Month

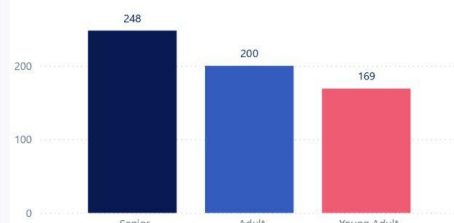
● Total Revenue ● Revenue Growth MoM %



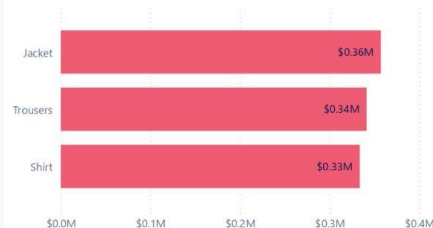
Purchase Frequency Distribution



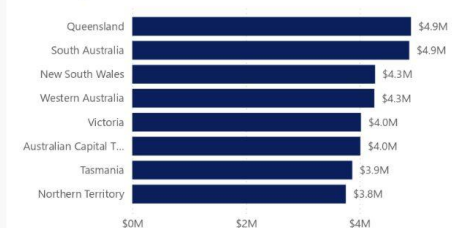
Customer Distribution by Age Group



Revenue by Product Type



Revenue by State



Revenue by Age Group



# Strategic Recommendations

Recommendations are prioritized based on customer reach, revenue contribution, and potential business impact.

## Loyalty Program



**Insight Source** Repeat Purchase Rate: 26.3%

### **Objective:**

- ✓ Increase repeat purchases

### **Expected Impact:**

- ✓ Improve customer retention and customer lifetime value (CLV)

## Senior Customer Campaign



**Insight Source** Senior Revenue: \$13.4M

### **Objective**

- ✓ Retain high-value customers

### **Expected Impact**

- ✓ Maximize revenue from the most valuable segment

## Regional Optimization



**Insight Source** Queensland Revenue: \$4.9M

### **Objective**

- ✓ Strengthen high-performing regions

### **Expected Impact**

- ✓ Increase revenue contribution from Queensland and other top states

## Product Focus Strategy



**Insight Source** Jacket Revenue: \$0.36M

### **Objective**

- ✓ Prioritize top-performing products

### **Expected Impact**

- ✓ Improve inventory efficiency and product revenue

1

## Loyalty Program



Highest Impact

2

## Senior Customer Campaign



High Impact

3

## Regional Optimization



Medium-High Impact

4

## Product Focus Strategy



Medium Impact

# Conclusion

## 1. Revenue Performance

Revenue reached

**\$33.97M**

with positive growth of

**8.9%**



## 2. Retention Opportunity

Customer retention remains the largest opportunity for sustainable growth.



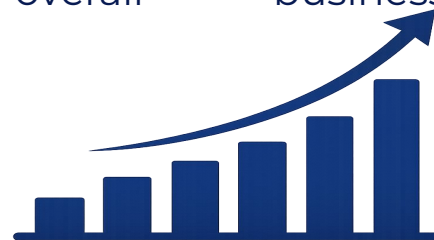
## 3. Key Revenue Drivers

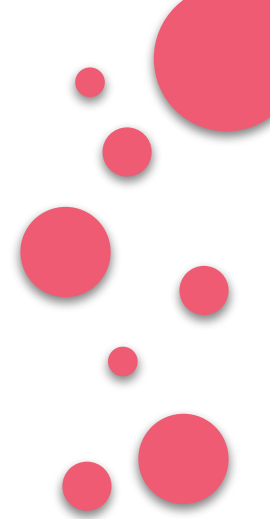
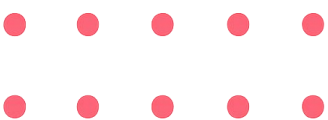
**Senior** customers, **Jackets**, and **Queensland** emerged as the strongest contributors to overall revenue performance.



## Final Conclusion

Improving **customer retention** while focusing on **high-value customer segments, top-performing products, and key regions** will support sustainable revenue growth and strengthen overall business performance.





**THANK  
YOU!!**



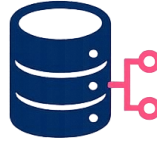
# Appendix

A1



**Root Cause  
Analysis**

A2



**ERD & Data  
Structure**

A3



**Data Validation  
Summary**

A4



**Key SQL Query**

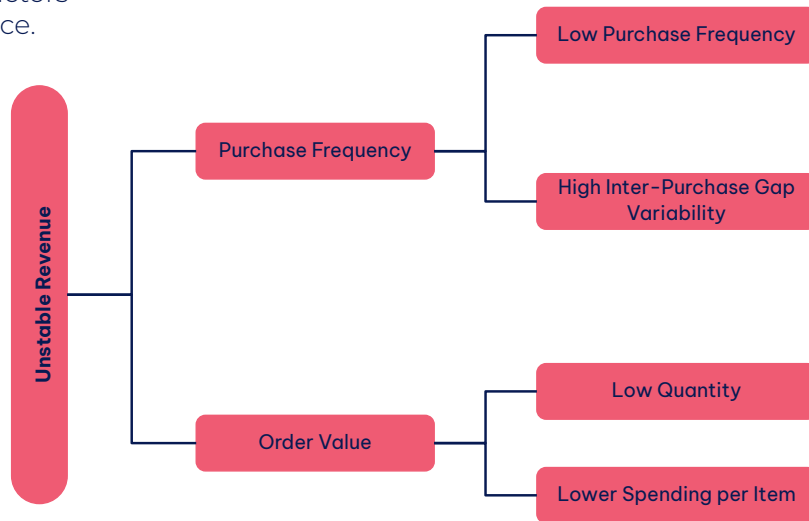
A5



**Business Metrics  
Definition**

# A1 Root Cause Analysis

Framework developed to identify factors contributing to revenue performance.



Connection to Current Analysis

## Purchase Frequency

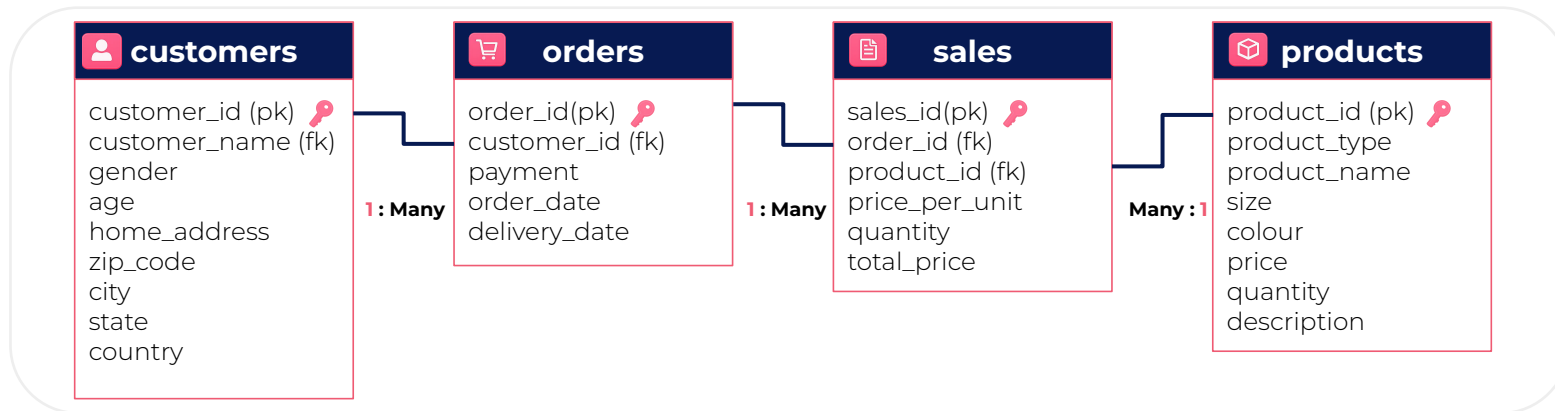
→ Analyzed through Repeat Rate and Customer Retention

## Order Value

→ Analyzed through Revenue Contribution by Customer Segment, Product, and Region



# A2 ERD & Data Structure



## Key Tables



### Customers

Stores customer demographic and geographic information.



### Orders

Stores purchase transactions and order dates.



### Sales

Contains transaction-level product purchases and revenue details.



### Products

Contains product attributes and inventory information.

## Data Relationships

1

### Customers → Orders

One customer can place multiple orders.

2

### Orders → Sales

One order can contain multiple purchased products.

3

### Products → Sales

One product can appear in multiple sales transactions.

4



The relationships enable customer, product, and revenue analysis.

# A3

## Data Validation Summary



### Missing Values

No missing values found in key columns.



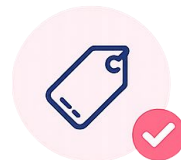
### Duplicate Records

No duplicate primary keys detected.



### Invalid Transactions

No invalid dates or negative values found.



### Price Consistency

No price inconsistencies identified



### Outliers Review

No significant outliers in transaction values.



The dataset passed all validation checks and was considered suitable for customer behavior, revenue, and segmentation analysis.

# A4 Key SQL Queries

## SQL #1

### Repeat Rate Analysis

```
WITH customer_frequency AS (  
  SELECT  
    customer_id,  
    COUNT(order_id) AS purchase_frequency  
  FROM orders  
  GROUP BY 1  
)
```

```
SELECT  
  purchase_frequency,  
  COUNT(customer_id) AS total_customers  
FROM customer_frequency  
GROUP BY 1  
ORDER BY 1;
```



#### Insight Generated

Repeat Rate  
**26.3%**  
of customers made  
repeat purchases.

## SQL #2

### Revenue by Age Group

```
SELECT  
CASE  
  WHEN age BETWEEN 20 AND 35 THEN 'Young Adult'  
  WHEN age BETWEEN 36 AND 55 THEN 'Adult'  
  ELSE 'Senior'  
END AS age_group,  
SUM(payment) AS total_revenue  
FROM orders o  
JOIN customers c  
ON o.customer_id = c.customer_id  
GROUP BY 1;
```



#### Insight Generated

Senior customers  
contributed the highest  
revenue  
**(\$13.4M)**

## SQL #3

### Revenue by Product Type

```
SELECT  
  product_type,  
  SUM(total_price) AS total_revenue  
FROM sales s  
JOIN products p  
ON s.product_id = p.product_id  
GROUP BY 1  
ORDER BY 2 DESC;
```



#### Insight Generated

**Jacket**  
generated the highest  
revenue  
**(\$0.36M)**

## SQL #4

### Revenue by State

```
SELECT  
  state,  
  SUM(payment) AS total_revenue  
FROM orders o  
JOIN customers c  
ON o.customer_id = c.customer_id  
GROUP BY 1  
ORDER BY 2 DESC;
```



#### Insight Generated

**Queensland**  
generated the highest  
revenue  
**(\$4.9M)**

# A5

# Business Metrics Definition

1



Total Revenue

## FORMULA

$\text{SUM (payment)}$

## PURPOSE

Measures total revenue generated during the analysis period.

2



Average Order Value (AOV)

## FORMULA

$\frac{\text{Total Revenue}}{\text{Total Orders}}$

## PURPOSE

Measures the average spending per transaction.

3



Repeat Rate

## FORMULA

$\frac{\text{Repeat Customers}}{\text{Total Customers}}$

## PURPOSE

Measures the percentage of customers who made more than one purchase.

4



Revenue Growth %

## FORMULA

$\frac{(\text{Current Revenue} - \text{Previous Revenue})}{\text{Previous Revenue}}$

## PURPOSE

Measures revenue performance changes over time.